

CHAPTER ONE: INTRODUCTION

1.1 TITLE

Design and development of a Vision-Based Pinhole Detection system for Blown Film Extrusion Lines of food packaging materials.

1.2 BACKGROUND

Blown film extrusion is widely used for manufacturing plastic films for food packaging. Pinholes can lead to contamination, spoilage and reduced shelf life of packaged goods. In food applications, such defects are unacceptable. This project proposes the integration of a real-time optical inspection system into a blown film line to detect pinholes in polymer films ranging from 15-90 micron in thickness.

This project was inspired by a real industrial challenge I encountered during my internship at PE Plastics, a company that manufactures food packaging material using blown film extrusion. One of the extruders consistently produced rolls with undetected pinholes. These pinholes were invisible during the extrusion process and their presence was only discovered during downstream use particularly when hot liquids leaked through the pinholes during auto-pack operations.

The auto-pack was for 'mahewu', mahewu is packed whilst hot. When hot liquid was dispensed into the packaging, it would leak or gush out through these tiny pinholes, resulting in significant product loss, compromised hygiene and customer dissatisfaction. Due to the absence of an in-line detection system capable of identifying such micro-defects in real time, the only short-term solution was to completely stop using that particular extruder, leading to down time and production inefficiencies.

This experience highlighted the critical importance of early pinhole detection in food packaging applications, where even microscopic defects can cause major quality and safety issues. It also

revealed a gap in process automation and quality control, particularly in environments where visual inspection is impractical or unreliable due to high-speed production.

Motivated by this experience, I decided to focus my HIT400 project on developing a low-cost vision-based pinhole detection system for blown film extrusion. The goal is to create a solution that can be integrated directly into the production line, capable of identifying pinholes in real time before the film reaches the winding stage. By addressing this problem, the project aims to enhance product reliability, minimize waste and improve overall quality assurance in the manufacturing of food packaging materials.

1.3 PROBLEM STATEMENT

Current quality control in blown film extrusion largely relies on manual visual inspection or offline testing methods that fail to identify pinholes during production. This delay results in waste, production downtime and the potential distribution of defective food packaging materials. Existing automated detection technologies are often expensive, proprietary and inaccessible for small to medium manufacturers or academic applications.

There is therefore a critical need for an affordable, open-source and real-time vision-based system capable of detecting microscopic pinholes in thin (15-90 μ m) polyethylene films during extrusion.

1.4 AIM

To design and implement a camera-based vision system capable of detecting pinholes during production, ensuring the integrity and safety of food packaging materials in real time.

1.5 OBJECTIVES

- To design a low-cost, non-contact optical detection system using a high-resolution camera and backlight to identify pinholes during production.
 - To develop image processing software capable of detecting, classifying and recording pinhole defects in real-time.
 - To test and evaluate the effectiveness of the detection system under controlled conditions using film samples with known defects.
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1.6 JUSTIFICATION

The production of blown film for food packaging requires strict quality control because defects such as pinholes can compromise barrier properties, resulting in contamination, reduced shelf life, and potential health risks to consumers. Pinhole defects may arise from gel formation, melt instability, contamination, or mechanical damage during processing (Ding, Hobbs & Bhattacharya, 2014). In food packaging applications, even microscopic perforations can allow oxygen or microorganisms to penetrate the product, undermining food safety systems. Therefore, implementing an automated, camera-based pinhole detection system is essential to ensure consistent quality, reduce material waste, and prevent defective products from reaching the market.

In Zimbabwe, food safety management systems are guided by national standards such as **ZWS 749: Requirements for a Hazard Analysis and Critical Control Point (HACCP) system**, which outlines preventive approaches to identifying and controlling hazards within the food supply chain (Standards Association of Zimbabwe, 2024). Since packaging materials form part of the food safety chain, defects in polymer films can represent a potential hazard if not adequately controlled. The Standards Association of Zimbabwe (SAZ) also provides accredited certification services aligned with HACCP and ISO-based food safety systems (Standards Association of Zimbabwe, no date). By integrating a vision-based defect detection system into the blown film extrusion line, this project supports HACCP principles by introducing a

preventive monitoring mechanism at a critical control stage. Furthermore, vision-based inspection technologies have been shown to significantly improve defect detection accuracy and reliability in polymer film manufacturing (Rawashdeh, Hazaveh & Altarazi, 2023; Szeliski, 2022). Consequently, this research is necessary to enhance food packaging safety, ensure compliance with Zimbabwean regulatory standards, and improve overall manufacturing efficiency and product integrity.

1.7 SCOPE OF STUDY

This study focuses on the design, development and evaluation of a pinhole detection system for blown film extrusion lines producing food-grade polymer films ranging from 15 to 90 microns in thickness. The research is confined to the implementation of an optical inspection system using image-based technology for real-time detection of pinholes during the manufacturing process.

- I) Identification of causes of pinholes
Investigating common sources of pinhole defects in blown film, including raw material impurities, melt inconsistency, air contamination and process disturbances.
- II) Selection of suitable detection technology
Evaluating and selecting appropriate optical components such as industrial cameras and lighting systems capable of detecting micron-level defects in transparent or semi-transparent films.
- III) System design and integration
Designing a mounting and alignment system for camera and backlight integration into the existing blown film line, preferably in the flat film zone (post-nip rollers)
- IV) Image processing and analysis
Developing or applying software algorithms (e.g. thresholding, blob detection) to identify and quantify pinholes in real-time.

V) Prototype testing and calibration

Testing the system using known pinhole sizes and real production samples to assess accuracy, sensitivity and reliability.

VI) Evaluation of effectiveness

Measuring the system's performances in terms of

- ✓ Minimum detectable hole size
- ✓ False positives or negatives
- ✓ Speed compatibility with line throughput

VII) Limitations and industrial scalability

Assessing constraints such as lighting uniformly, speed limitations and film movement and suggesting improvement for future industrial-scale applications.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Blown film extrusion is widely used to manufacture polymer films for packaging, including food packaging where barrier integrity is critical. A blown film line extrudes a polymer tube which is inflated to form a bubble, cooled and flattened into rolls. Defects such as pinholes, gels, contamination specks, gauge variation, haze, die lines and melt fracture degrade performance and may compromise barrier properties, product appearance and food safety. Detecting pinholes (microscopic through-holes or perforations) is particularly important in food packaging because even tiny pinholes can permit oxygen or moisture ingress that shortens shelf life or causes contamination. Online detection that is accurate at high line speeds is therefore essential for quality control. This review focuses on how pinholes and other defects have been detected on blown film extrusion lines, surveying both manual/visual inspection and automated, vision—based and sensor-based systems.

2.2 Blown film extrusion: process overview and defect taxonomy

a) **Process summary**

The blown film process melts polymer resin and extrudes it through a circular die. The extruded tube is inflated via annular air ring, cooled across a frost line, collapsed by nip rollers, and wound. Process variables include resin grade, melt temperature profile, die design, cooling rate, blow-up ratio, nip roll settings, and winding tension, each of which can influence defect formation. (“HOW TO SOLVE BLOWN FILM PROBLEMS,” n.d.)

b) **Defect types (typology and significance)**

Common defects in blown film include:

- Pinholes/micro-holes: through-film perforations (critical for barrier films).
- Gels: un-melted resin particles causing visual blemishes and weak spots
- Contamination specks: foreign particles causing visible dark specks or thin spots.
- Gauge/thickness variation: nonuniform film thickness that affects

mechanical properties.

- Haze/cloudiness: optical defects affecting appearance.
- Die lines/sharkskin/melt fracture: surface irregularities from rheological instabilities.
- Burn marks, black specks: thermal degradation or char particles. (“HOW TO SOLVE BLOWN FILM PROBLEMS,” n.d.)

Each defect class has different detection challenges: pinholes require transmission detection to detect light passing through; specks and gels are often visible in reflection; gauge variation benefits from thickness metrology sensors; haze requires optical characterization. (Rawashdeh et al., 2023)

1. Manual (human) inspection vs automated inspection

a) Manual inspection

Historically, quality control relied on operator visual inspection at roll change off-line checks (e.g., feel, light table inspection) and occasional laboratory tests. Manual methods are subjective, fatiguing, and incompatible with high line speeds and the need for consistent detection thresholds. Reports and industry guides emphasize the limitations of manual inspection for micro-defects and continuous process control. (“HOW TO SOLVE BLOWN FILM PROBLEMS,” n.d.)

b) Automated inspection: motivations and benefits

Automated systems provide consistent detection thresholds, logging, high-throughput screening and integration with process control. Automated inspection also enables 100% in-line inspection (scan every meter of film), rapid rejection of defective batches, and feedback for process tuning. For pinholes in particular, automated transmission inspection systems provide detection sensitivity beyond human capability at production speeds. (“pinhole inspection units,” n.d.)

2.2.2 Sensing modalities for pinhole and general defect detection

Detection approaches fall into several sensor/ illumination categories:

a) Transmission/backlight (full-field or line) imaging

Backlight (transmission) inspection places a uniform light source behind the film and images

the film's transmitted light. Pinholes and thin areas allow more light through and appear as bright spots in transmitted images. Transmission is the most common and effective method for pinhole detection because it directly measures through-film light passage; manufacturer units and commercial pinholes detectors use carefully controlled low-noise photonic sensors and ring/backlight illumination to detect micro-perforations as small as 1µm under optimal conditions.

Key design considerations: uniform illumination, sensor dynamic range, optical resolution versus line speed, stray light suppression and wide field coverage (to inspect the entire film width).

High sensitivity cameras or photodiode arrays are often used.

b) Low-light/phonics detectors and high dynamic range sensors

Some systems leverage photomultiplier or low-light CMOS sensors with high gain to detect very small pinholes, especially under low ambient light and for metallized or opaque films where contrast is low.

c) Reflective vision and inspection (for surface defects)

Reflective imaging with structured illumination or grazing incidence is better suited for surface defects such as gels, contamination or die lines. Combined systems using both reflective and transmissive imaging can detect a broader and set of defects.

d) Thermography and IR

Thermographic imaging (e.g., active thermography) reveals subsurface defects by heating and observing thermal response; some studies apply thermography with ANN classification for pinhole or subsurface void detection in films/coatings. Thermography tends to be used in research or specific industrial contexts rather than standard blown film lines because of complexity and slower acquisition.

e) Densitometry/transmission densitometers

Full-field transmission densitometry measures transmitted light intensity across the film and has been used to detect thin spots and pinholes in cast or metallized films; it may be combined with machine vision for localization.

f) Electrical or leak testing

For finished packages, leak testing (e.g., pressure decay, bubble testing in liquid baths, conductive tests) is common; however, these are typically offline or after forming (e.g., blister packs) and less suitable for continuous blown film inspection prior to conversion.

2.2.3 Image acquisition & illumination engineering

Accurate pinhole detection depends more on acquisition optics and lighting than on classification algorithms alone. Key factors documented in industry and research sources include:

- Backlight uniformity: uneven backlight creates false positives/negatives. Manufacturers design diffusers, collimated LED arrays and integrating cavities for uniformity.
- Spectral selection: choosing wavelengths where the film attenuates helps increase contrast; for metallized films near-IR or specific narrowband lighting may improve detection.
- Sensor noise and dynamic range: high dynamic range is necessary to detect tiny transmission differences without saturation at bright regions; some vendors use dedicated low-noise sensors.
- Optical resolution: optics must resolve the smallest defect size of interest; Nyquist sampling considerations dictate pixel pitch and lens quality.
- Industrial pinhole detection modules from Hamamatsu and Antares describe engineered illumination and sensor packages optimized for these needs.

2.2.4 Image processing and machine learning methods

a) Traditional image processing

Early systems used straightforward image processing pipelines: flat-field correction, background subtraction, adaptive thresholding, morphological filtering, connected component analysis, and heuristic size/shape filters to eliminate noise. For transmission images, simple thresholding on transmitted intensity identifies candidate pinholes which are then morphologically processed to remove speckle noise. These methods work well when lighting is uniform and defect contrast is high.

b) Statistical and transform methods

Fourier and wavelet analyses, texture measures, and statistical filters have been used to separate defect structures from patterned texture. For certain film textures, frequency filtering helps suppress periodic noise (e.g., weave patterns of films) and reveal localized defects.

c) Machine learning and neural networks

More recent work applies machine learning and deep learning to reduce false positives and classify defect types. Convolutional neural networks (CNNs) and hybrid architectures have been trained (often on labeled datasets) to recognize pinholes, gels and contamination. Several academic studies and industry systems use ML for post-processing after traditional detection to cluster and classify defect candidates, improving precision at the cost of requiring labeled training data. A 2023 vision-based online defect detection study demonstrated ML integration for polymeric films and argued for real-time feasibility.

d) Hybrid approaches

Best practice is to combine simple, fast image processing (for initial candidate detection) with ML classification to filter false positives. This limits computation load and enables real-time performance at industrial speeds. Systems may also employ change detection and temporal voting across consecutive frames to confirm transient defects versus noise.

2. Causes of pinholes

Multiple consistent root causes for pinholes are reported across patents, manufacturer notes and technical literature:

- Contaminants/particles: dust or foreign particles entrained in the melt or landing on film during winding can create pinholes (particle exclusion zones or pick-off).
- Volatile substances / outgassing: residual solvents, moisture or volatile additives can vaporize during extrusion or later thermal processes, causing bubbles that burst and leave pinholes. Patents and technical notes emphasize volatile entrainment as a key mechanism.
- Poor dispersion or un-melted resin: incomplete melting or poor dispersion can create weak spots and voids that develop into pinholes.
- Winding / re-winding adhesion (pick-off): metal transfer or adhesion issues during rewinding may pull small pieces away and create pinholes, especially in metallized films.
- Temperature control issues: improper heater zone settings, inadequate mixing, or too-low melt temperatures cause poor melt quality and pinholing. Industry troubleshooting guidance lists heater profile and screen damage as common causes.

Because causes are varied, prevention requires good material handling, melt filtration,

controlled drying to remove moisture/volatile content, correct temperature profiles and maintenance of filtering/screens.

2.3 Detection of other defects and their detection methods

- Gels and contamination: reflective imaging with controlled lighting and high resolution is effective; algorithms use contrast enhancement and blob detection.
- Gauge variation: thickness gauges (beta gauges, infrared thickness meters, laser triangulation) and full-field thickness mapping are used. Vision systems may infer thickness variations using structured light or transmission densitometry.
- Haze/optical quality: haze meters and optical scatter methods quantify haze; vision systems estimate image contrast and scatter metrics.
- Die lines / melt fracture: high-resolution line-scan cameras with reflective imaging detect linear features; temporal analysis tracks repeatability.

Combining multiple sensor types extends detection coverage across defect classes. Several vendors provide modular systems that combine backlight and reflective cameras to simultaneously find pinholes and surface defects.

2.4 Commercial solutions and industrial practice

A number of vendors provide dedicated pinhole inspection hardware or full film inspection systems for converting lines:

Hamamatsu: manufactures pinhole inspection units and low-light photonics units claiming detection down to micrometer scales, suitable for roll-to-roll inspection at high speed.

Antares Vision Group: offers pinhole detection and inspection systems for laminated films and pharmaceutical packaging; integrates AI and industrial sensors.

Erhardt+Leimer: SMARTSCAN 200% inspection with pinhole detection for converting industry.

Others: specialized suppliers (Jornen, Patera/Scanware, QUALISHIELD by ACG) supply both off-the-shelf and custom inspection units targeted to blister packs, foil, and film converting.

Industrial systems emphasize integration with line control (e.g., stop/fail signals, reject triggers) and operate at typical converting speeds with real-time throughput. Performance

claims vary by vendor.

2.5 Performance metrics and challenges

a) Accuracy, sensitivity, and false positives

Sensitivity: detecting very small pinholes (sub-10 μm) requires high optical resolution and high signal-to-noise ratio; vendor claims vary (some claim 1 μm under ideal conditions).

False positives: dust, specular highlights, wrinkles and winding defects can generate false alarms; post-processing, temporal confirmation and ML classification are used to reduce false alarms.

b) Throughput and real-time constraints

At industrial line speeds (meters per minute to hundreds of m/min), image readout, processing and I/O must meet tight latency budgets. Line-scan cameras or parallel multi-camera arrays are common solutions; architectures that perform fast initial filtering in FPGA or edge devices followed by more sophisticated analysis on CPU/GPU are used.

c) Material variability and environmental conditions

Differences in film thickness, opacity, colors, metallization, and winding tension lead to variability requiring robust calibration and adaptive algorithms. Ambient light and vibration on lines create additional noise.

d) Data labeling and model generalization

Machine-learning approaches need labeled datasets that cover the diversity of defect appearances and production conditions. Gathering large, balanced datasets for rare defects like pinholes is challenging; synthetic augmentation and transfer learning are strategies used in research.

2.6 Gaps identified in the literature & promising directions

From the sources reviewed, the following gaps and research opportunities are apparent:

- I) Limited open datasets for pinhole detection — many vendors claim lack independent published datasets or benchmarks. Public datasets would accelerate algorithmic research.
- II) Hybrid multi-sensor fusion — while combinations of backlight and reflective

imaging are used in industry, academic literature on optimal sensor fusion architectures for blown film (balancing cost, speed and detection performance) is limited.

- III) Real-time ML deployment on constrained hardware — more work is needed on deploying accurate ML classifiers on low-latency embedded hardware or FPGA systems tailored to line-speed constraints.
 - IV) Robustness to metallized films and opaque layers — specialized illumination/spectral strategies for metallized films need broader documentation and comparative studies.
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2.7 Recommendations for designing a vision-based pinhole detection system for blown film lines

Based on the literature and industrial practice, the following recommendations are supported:

- a) Primary modality: transmission/backlight imaging — implement a high-uniformity backlight across the full web width with low-noise, high-dynamic range sensors for primary pinhole detection. Use narrowband LEDs if spectral tuning helps contrast.
- b) Dual modality — add a reflective camera bank for surface defects (gels, scratches) so a single inspection station detects both pinholes and surface anomalies.
- c) Two-stage processing pipeline — stage 1: fast thresholding and morphological filtering on FPGA/edge device to detect candidates; stage 2: ML classifier (CNN) to reduce false positives and classify defect types. Use temporal voting across frames for transient noise rejection.
- d) Calibration, adaptive thresholds and logging — implement routine flat-field calibration, auto-gain control, and per-run baseline learning to accommodate material variability. Provide full logging and visualization so operators can review candidate defects and perform root-cause analysis.
- e) Preventative measures — pair detection with process control and preventative maintenance: melt filtration, drying procedures, screen checks, and controlled winding tension can reduce pinhole incidence. Use detection feedback to flag runs for process intervention rather than only roll rejection.
- f) Validation and acceptance testing — establish test artifacts with calibrated pinhole sizes (targets) to validate sensitivity claims and set operating thresholds; measure

detection performance (precision, recall) and line throughput under production conditions.

2.8 Conclusion

The literature and industry materials converge on transmission/backlight imaging as the foundation for pinhole detection on blown film lines, often supplemented by low-light sensors and dedicated photonics modules for very small defects. Traditional image processing techniques remain useful for initial detection; machine-learning classifiers increasingly help reduce false positives and extend detection to multiple defect classes. Commercial vendors provide turnkey solutions, but independent evaluations and open datasets are rare. Designing a robust vision-based pinhole detector requires careful attention to illumination engineering, sensor selection, real-time processing architecture, and process integration (prevention + detection). Future research should prioritize sensor fusion, deployment of ML in constrained real-time environments, and standardized datasets/benchmarks for pinhole and film defects.

CHAPTER THREE: METHODOLOGY

3.1 INTRODUCTION

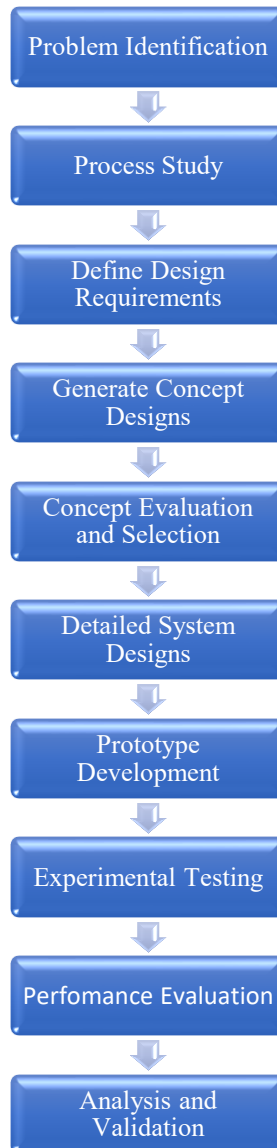
This chapter presents the methodology adopted for the design, development, and validation of a camera-based pinhole detection system for blown film extrusion lines used in food packaging production. The purpose of this chapter is to describe, in a structured and reproducible manner, the engineering approach followed to transform the identified industrial problem into a functional prototype capable of real-time defect detection.

The chapter outlines:

- The research design approach
- Analysis of the blown film extrusion process
- Development of multiple conceptual equipment designs
- Evaluation and selection of the optimal concept
- Hardware and software architecture
- Image acquisition strategy
- Algorithm development
- Experimental validation procedure
- Performance evaluation methods
- Safety considerations and system limitations

The methodology ensures that the system is technically feasible, economically viable, safe for industrial use, and aligned with food packaging quality requirements. This chapter further outlines how the proposed system integrates hardware and software components to enable real-time detection of pinhole defects in polymer film before winding.

3.1.2 Overall Methodology Framework



3.2 Research Design

This study adopts an **applied experimental engineering research design**. The goal is not only theoretical investigation but also the development of a practical working prototype.

3.2.1 Nature of the Research

- Applied research
- Experimental validation
- Prototype-based development

- Quantitative system evaluation

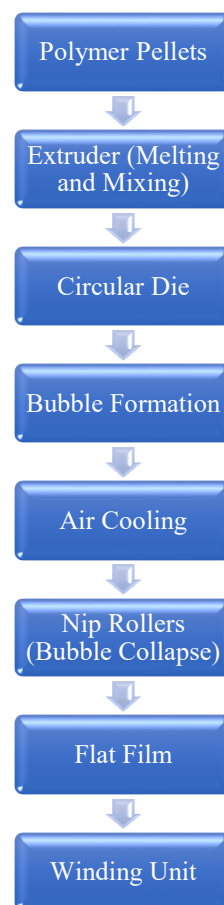
The research integrates:

- Mechanical design
 - Optical engineering
 - Image processing
 - Polymer processing knowledge
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3.3 Description of the Blown Film Extrusion Process

Understanding the extrusion process is critical before inserting an inspection system. Blown film extrusion consists of polymer melting, bubble formation, cooling, collapsing and winding.

3.3.1 Process Flow



Polymer Pellets

This is the first stage of the process, raw material feeding happens here. Polymer pellets are small granules of thermoplastic material, commonly Low-Density Polyethylene (LDPE), Linear Low-Density Polyethylene (LLDPE), High-Density Polyethylene (HDPE) and Polypropylene (PP). for food packaging, materials must comply with food safety standards. For this research I will be focussing on packaging material made from LLDPE and LDPE. The composition of the material I will use as a sample is 80% medium duty LDPE + 20% LLDPE. The raw materials are poured into a mixer according to the composition ratios mentioned above, additives such as slip agents, anti-block agents, UV stabilizers and colour masterbatch may be added and mixed well with the polymer pellets in the mixer. The mixture is passed through a dryer to remove moisture. The pellets are loaded into a hopper. This stage is important because if raw materials contain moisture, it will cause bubbles, voids and pinholes, any present contaminants cause weak spots and film defects and poor mixing will lead to thickness inconsistency. Defects originating at this stage can cause pinholes later in the film.

Extruder (melting and mixing)

The main components are barrel, rotating screw, heaters and temperature controllers, motor and gearbox. The barrel has three zones, i.e., feeding zone, compression zone and metering zone. The screw rotates and works in all three zones.

- a) Feeding Zone: pellets enter from the hopper and the material begins to compact. Friction begins heating the material.
- b) Compression Zone: channel depth decreases, polymer begins to melt and pressure builds up. Air pockets are removed at this zone.
- c) Metering Zone: the polymer is now fully molten, it has uniform temperature, stable pressure and steady flow toward the die.

The critical parameters are barrel temperature, screw speed, pressure stability and melt homogeneity. These matter because if temperature is too low there will be incomplete melting which causes gel defects, if the temperature is too high, the polymer degrades and pressure fluctuations cause thickness variation. All these can cause weak film areas and pinholes.

Circular die

Before the melt reaches the die, it passes through fine mesh screens where contaminants are filtered out. The melt flow becomes more uniform. The importance of this is that it removes un-melted particles, removes dirt or degraded polymer and prevents die blockage. Without

filtration foreign particles can cause micro-tears, weak film zones and pinhole formation. The film is formed at this stage; the molten polymer becomes a tube. The circular die distributes molten polymer evenly around a ring. The polymer melt exists as a thin tubular shape. Wall thickness is controlled by die gap, melt pressure and flow uniformity. The critical die design factors are die gap precision, flow channel symmetry and temperature uniformity. The die is the common point of origin for pinholes. Other possible defects at the die are uneven flow which leads to thickness variation, die lip damage which causes streaks and poor pressure control which leads to weak spots.

Bubble formation

Air is introduced inside the tube. The internal air pressure inflates the tube; film diameter increases and stretching occurs in machine direction (MD) and transverse direction (TD). This improves mechanical properties. If air pressure fluctuates, bubble instability occurs, if stretching is uneven, it forms weak zones and over-stretching leads to thinning and causes pinholes. Blow up ratio is critical at this point; it is given by the formula $BUR = \frac{\text{bubble diameter}}{\text{die diameter}}$

Air cooling

The air ring is responsible for cooling the bubble immediately after exiting the die. Air is blown around the bubble, polymer cools and solidifies and film becomes stable. The key parameters are air temperature, air velocity and uniform distribution. Problems that may arise are if uneven cooling happens there is thickness variation, presence of turbulence causes bubble instability and poor cooling will result in a weak film structure. Uneven crystallization reduces mechanical strength.

Nip rollers

Before the tube reaches the collapsing frame there is a frost line, where the film becomes fully solid. The polymer transitions from molten to solid and molecular orientation is locked in. if the frost line is too high there is slow cooling, if it is too low there is rapid cooling stress and inconsistent frost line cause mechanical weakness. The collapsing frame is where the bubble flattens. It is guided into a flat sheet. Guides and frames collapse the tube, air inside is removed and tube becomes a double-layer flat film. The risks are misalignment and it causes wrinkles, excess tension causes tearing and mechanical stress causes micro-cracks. The nip rollers pull film upwards, maintain tension and control film thickness. Film thickness depends on screw speed, nip speed and air pressure. If nip speed increases the film becomes thinner. There is an increased risk of pinholes if the film is too thin.

Flat film

There is a flat film after the nip rollers, this is where the detection system will be integrated. This position is ideal because film fully solid, film is flat, there is no bubble distortion and the surface is stable for imaging. This proves it is the most reliable inspection point.

Winding unit

This is the final stage. Film tension is controlled, roll diameter increases gradually and the finished product is prepared for use. If defects are not detected before winding, entire roll becomes scrap, the cost increases and food safety risk increases.

3.3.2 Why inspect after nip rollers?

The inspection zone is placed **after the nip rollers and before winding** because:

- Film is flat and tensioned
- Minimal vibration
- Stable geometry
- Constant thickness
- Easy access for mounting camera and LED
- Does not interfere with bubble stability

This ensures reliable image capture.

3.4 Design Requirements

This section is about the requirements of the project, functional requirements, technical requirements and economic requirements. Functional requirements refer to what the system must do. These define the core purpose and operational capabilities of the system. Technical requirements refer to the engineering specifications necessary to achieve the functional requirements. They define how the system must be built in terms of hardware, software and performance capability. Economic requirements refer to the cost and financial constraints under which the system must operate. These ensure that the system is practical and viable for industrial implementation.

3.4.1 Functional Requirements

i. Real-Time Defect Detection

The system must detect pinholes while the film is moving during production, without stopping the extrusion process. This is important because stopping production increases downtime and real-time detection prevents defective rolls from being fully wound. This means, the camera must capture continuous images, the algorithm must process frames instantly, alerts must be immediate.

ii. Detection of Minimum Defect Size

The system must detect pinholes equal to or larger than a specified threshold (e.g., 100 μm), because very small pinholes can compromise barrier properties and food packaging safety depends on integrity. This requirement determines camera resolution, pixel-to-micron calibration and algorithm sensitivity.

iii. Non-Contact Operation

The system must not physically touch the film because contact can damage thin film, it may introduce contamination and it may interfere with film tension. Therefore, optical detection is required and no mechanical probes.

iv. Continuous Monitoring

The system must operate continuously during extrusion. This requires stable power supply, reliable lighting, robust software and automatic restart capability if errors occur.

v. Automatic Alert System

When a defect is detected, the system must trigger a visual or audible alarm, log the defect data and optionally mark the defective region. This ensures operator awareness, immediate corrective action and quality documentation.

vi. Data Logging and Traceability

The system must record time of defect, location on film, size of defect and image of defect. This supports HACCP compliance, quality audits and process improvement.

3.4.2 Technical Requirements

1. Camera Resolution

To detect a 100 μm defect each pixel represents a small enough physical dimension, resolution ≥ 5 MP is recommended and pixel-to-length calibration is required. If resolution is too low small defects will not be detected.

2. Frame Rate

The camera frame rate must match extrusion speed. If line speed = 60 m/min, camera must capture enough frames to avoid missing defects. The recommended frame rate is ≥ 60 fps. If there is low frame rate some defects will be missed.

3. Exposure Time

Exposure must be short to prevent motion blur. If film moves quickly an exposure of < 1 ms is preferred. If there is a blur it reduces detection accuracy.

4. Uniform Illumination

Lighting must be uniform across film width, have stable intensity, be flicker-free. Uneven lighting causes false positives and incorrect thresholding

5. Processing Speed

The image processing system must analyse each frame within milliseconds, operate without lag. Processing time per frame should be < 50 ms for real-time performance.

6. Mechanical Stability

Mounting structure must minimize vibration, maintain fixed focal distance and be adjustable. Vibration leads to image distortion.

7. Software Requirements

The system must include image acquisition module, pre-processing filters, thresholding algorithm, blob detection, measurement algorithm, decision logic, data storage module and the software must be robust and repeatable.

8. Calibration Requirement

The system must include physical calibration grid, pixel-to-millimetre conversion, periodic recalibration capability. Without calibration the measurements will be meaningless.

3.4.3 Economic Requirements

1. Affordable Initial Cost

The system should use commercial industrial camera (not research-grade ultra-expensive systems), standard LED panels and PC-based processing. The goal is to keep the total prototype cost reasonable for local industry.

2. Low Maintenance Cost

The system should use long-life LED lighting, avoid complex mechanical components and use software-based detection rather than mechanical sensors. This reduces long-term expenses.

3. Energy Efficiency

LED lighting must consume low power, produce minimal heat and operate continuously without overheating, energy efficiency reduces operational cost.

4. Scalability

The design should be easily upgraded, allow higher resolution cameras if needed and support future automation integration, this will increase industrial attractiveness.

5. Return on Investment (ROI)

The system should reduce material waste, prevent defective roll production, improve food safety compliance and reduce product recalls

The economy will benefit comes from reduced scrap, reduced rework, reduced food contamination risk and improved brand trust. These economic requirements ensure system is affordable, maintenance is manageable, energy cost is low, industrial adoption is realistic and long-term savings justify investment.

3.5 Conceptual Design Development

3.5.1 Introduction

The conceptual design development phase involves the generation, evaluation and selection of feasible design solutions that meet the identified system requirements. At this stage, abstract ideas are transformed into structured design concepts through systematic analysis, comparison of alternatives and consideration of technical, economic and operational constraints.

For the purposed vision-based pinhole detection system for blown film extrusion lines, this phase focuses on identifying suitable sensing methods, imaging configurations, illumination techniques, data processing algorithms and system integration strategies. Multiple design concepts are developed and assessed based on criteria such as detection accuracy, real-time performance, cost effectiveness, ease of implementation and compatibility with existing extrusion equipment.

Risk assessment is also conducted to identify potential technical limitations, including motion blur at high line speeds, lighting non-uniformity, false positives due to gels or contaminants and data latency issues. The selected conceptual design therefore represents the most technically feasible and economically viable configuration, forming the baseline for detailed mechanical design, algorithm development, hardware integration and prototype fabrication.

3.5.2 Components required

MECHANICAL COMPONENTS

- d) Structural frame: mild steel tubing, base support legs and adjustable height slots.
- e) Film handling system: rollers (aluminium idler roller, rubber-coated drive roller, stainless steel shafts and pillow block bearings), tension mechanism (spring-loaded tension roller)
- f) Drive system: speed controller, timing belt, dc geared motor, flexible shaft coupling
- g) Light isolation housing: enclosed inspection chamber, light baffles to eliminate ambient light, matte black internal panels

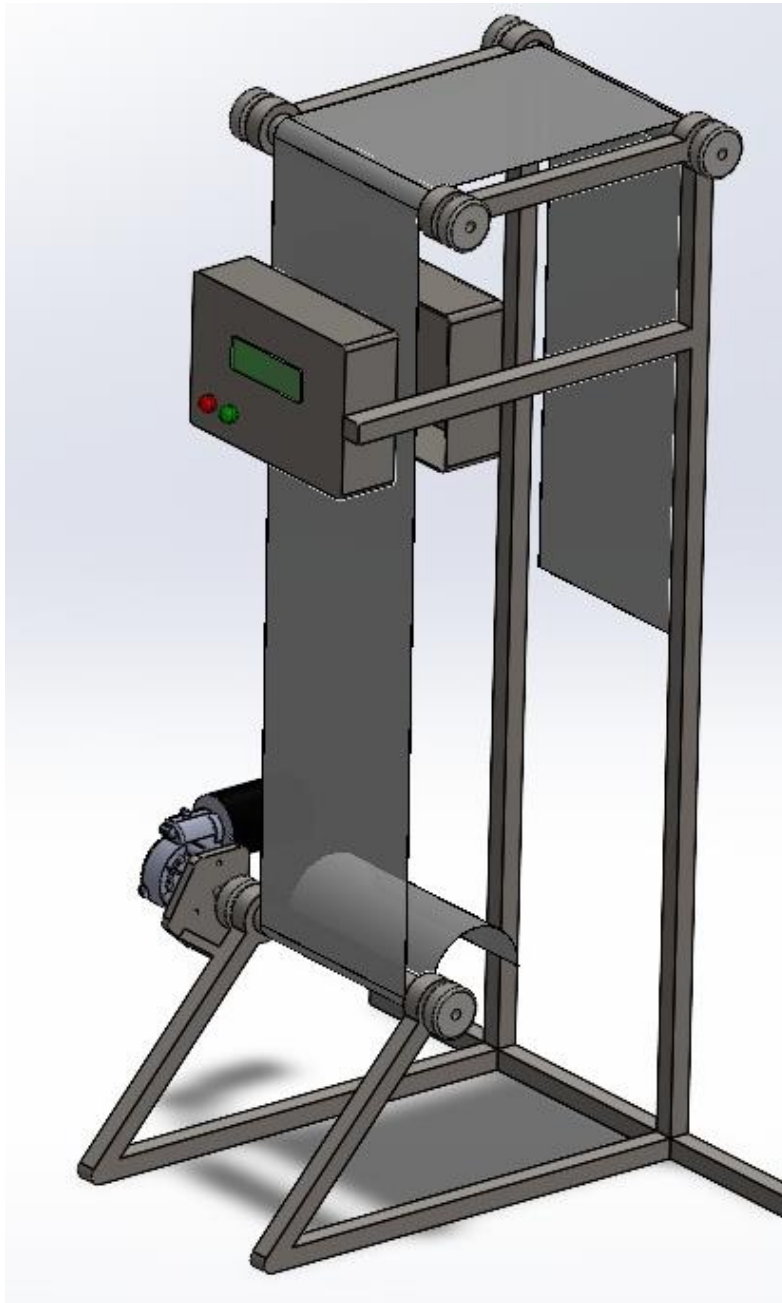
VISION AND ELECTRONIC COMPONENTS

- a) Camera system: ESPCAM
- b) Illumination system: bright light, lamp holder
- c) Control system: ESP32
- d) Sensors: emergency stop button, limit switches, rotary encoder and IP proximity sensor
- e) User interface: push button (start/stop), buzzer, LED indicator, display

SOFTWARE REQUIREMENTS

Thresholding algorithm, real-time defect counting, edge detection, python/C++

3.5.3 Conceptual Design

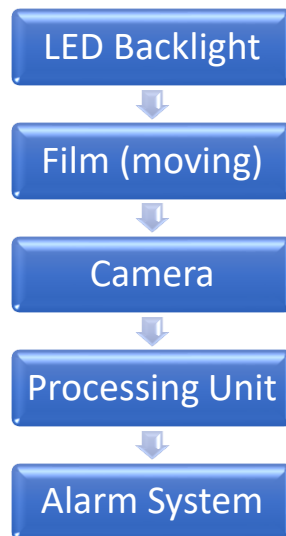




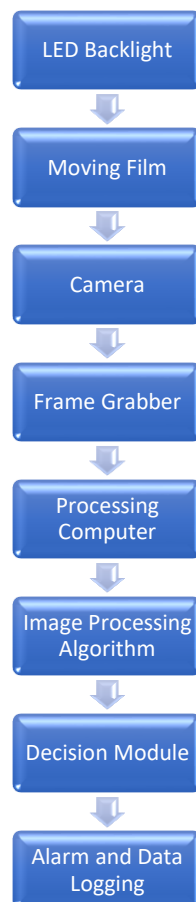
Working Principle:

The high-intensity LED backlight is placed on one side of the film, while a camera is placed on the opposite side. The film passes between the light source and the camera during production. The LED backlight produces a uniform illumination across the film surface, allowing the camera to capture continuous images of the moving film. If there are no defects, in this case pinholes, the polymer film partially blocks and diffuses the light, so the camera records a uniform grey image. When a pinhole defect is present the film is missing at that point. As a result, more light passes through the hole without being absorbed or scattered by the polymer. Therefore, the light does not get blocked by the pinhole, light passes through the hole more easily and the pinhole appears as a bright or lighter spot in the captured image.

The image processing algorithm then analyses the image by thresholding and defect detection techniques. Regions that are significantly brighter than the surrounding film are identified as potential pinhole defects. The system can then measure the size and location of the defect and trigger an alarm if the defect exceeds acceptable limit.



3.5.4 Final System Architecture



3.8 Hardware Implementation

Step 1: Mounting Structure

- Use aluminium frame
- Adjust brackets
- Anti-vibration mounts

Step 2: LED Installation

- Mount LED panel behind film
- Ensure uniform brightness
- Connect to stable power supply

Step 3: Camera Installation

- Align perpendicular to film
- Fix focal distance
- Adjust exposure

Step 4: Connect to Computer

- Use USB 3.0
 - Install camera drivers
-

3.9 Image Acquisition Method

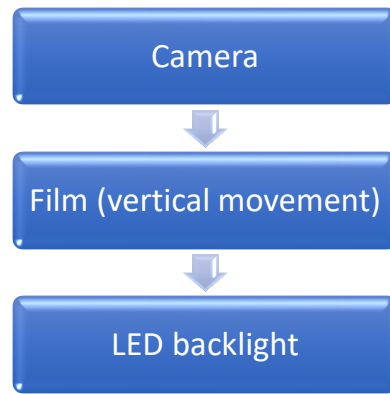
The key settings are:

Exposure time < 1ms (avoid blur)

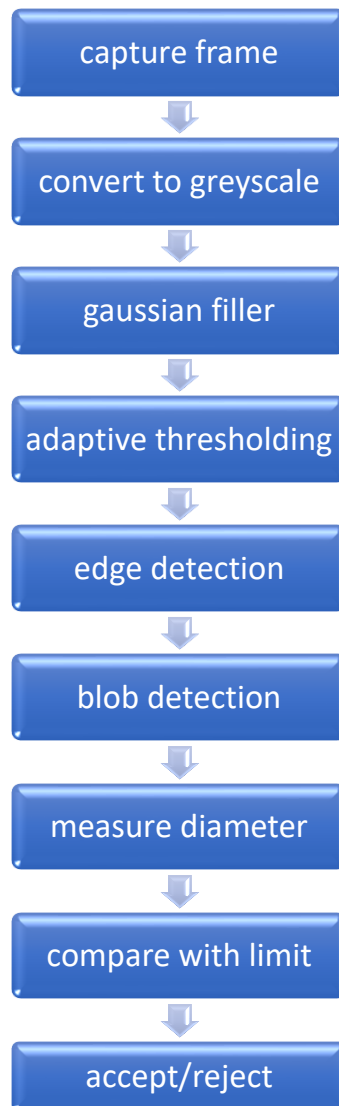
Fixed aperture

Constant lighting

Frame rate matched to line speed



3.10 Image Processing Algorithm



Mathematical Equations

Thresholding:

$$T(x, y) = 1 \text{ if } I(x, y) < \lambda$$

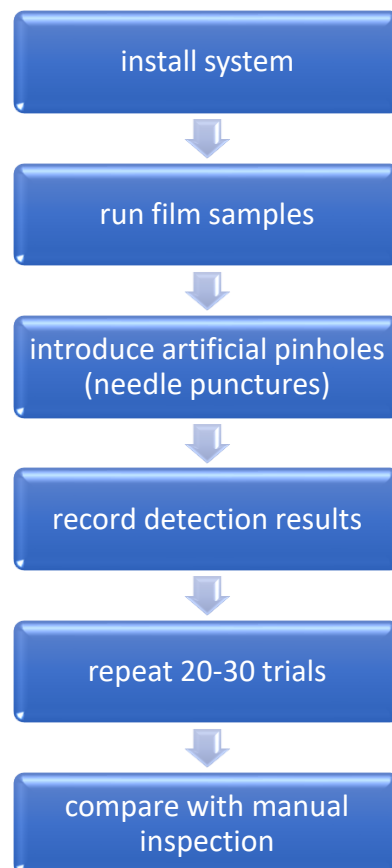
$$T(x, y) = 0 \text{ otherwise}$$

Equivalent diameter:

$$d = 2 \sqrt{\frac{Area}{\pi}}$$

If $D \geq 100\mu m$, defect detected

3.11 Experimental Procedure



3.12 Performance Evaluation

The performance of the system will be evaluated by testing the system using film samples containing artificial and naturally occurring pinhole defects. The evaluation will focus on the detection accuracy, detection speed and reliability of the system during continuous film movement. Parameters to evaluate include false detection rate, minimum detectable pinhole size, system response time, inspection coverage and detection accuracy.

The metrics that I will use are

1. Accuracy

$$accuracy = \frac{(TP + TN)}{total\ observations}$$

Where TP is true positives and TN is true negatives

2. Sensitivity

$$sensitivity = \frac{TP}{(TP + FN)}$$

Where FN is false negative

3. False positive rate

$$FPR = \frac{FP}{(FP + TN)}$$

Where FP is false positive

4. Processing time

Time per frame required is <50ms

3.13 Data Analysis

The captured data will be analyzed using statistical and image analysis techniques. Detected defects will be compared with manually measured pinhole sizes to determine detection accuracy. The results will be summarized using tables, graphs and statistical metrics such as detection rate and error percentage.

1. I will tabulate the results
 2. Using the table I will the graph of detection rate vs defect size
 3. I will compare manual inspection and automated inspection
 4. I will calculate the mean and standard deviation
-

3.14 Safety Considerations

- Insulated wiring
 - Secure mounting
 - No interference with rollers
 - Eye safety for LED
-

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